P29 graph-arm phased execution

Fixed public controls, 2026-10-22 PDT. Raw records: `bb8caa9722`; executable receipt: `3813ea4635`. Conductor T-4750 / T-4753.

This report describes implementation-control observations. It makes no discovery-stream yield or ECDLP cost inference.

Exact tested pair

Source: $\frac{E}{F_{320092037}}$: $y^2 = x^3 + 105931039x + 284015384$, prime order 320091991, $G = (0, 50656410)$. Vertex 8 after eight archived degree-7 separable maps over the same field: $y^2 = x^3 + 218000848x + 38636553$, transported $G = (148951155, 154566416)$. The source `low_x` base and its transported image have 256 x-orbits each; the fixed public targets use the same scalar coefficients.

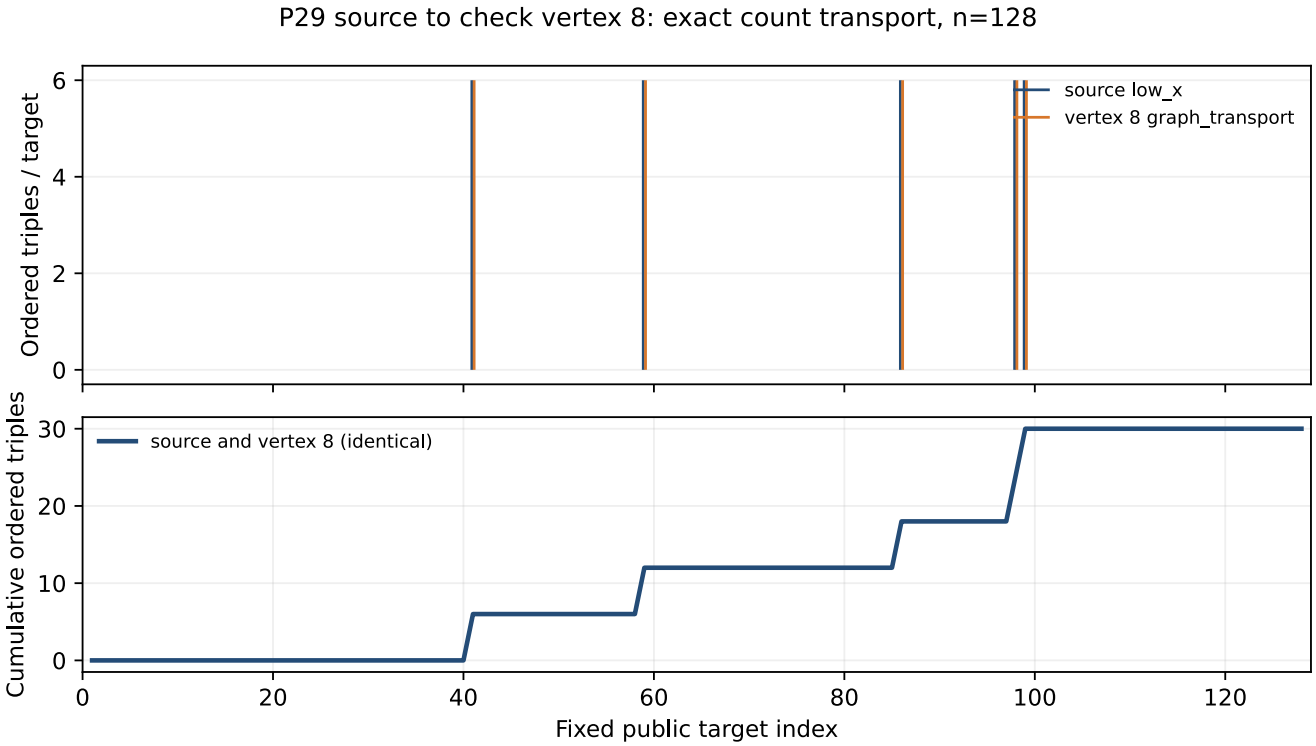


Figure 1: Exact source and vertex-8 ordered-count vectors. Both saved 128-target controls have five nonempty targets and 30 ordered triples; source and graph curves overlap exactly in this plot.

Recorded attempts and verification

Control 1 stopped before phases because checkout Python 3.13 lacked `psutil`. Control 2 stopped at input verification because the first wrapper invocation used positional arguments where flags were required. Both failure records remain saved. Control 3 passed with eight fixed targets and zero triples. Controls 4 (vertex 8) and 5 (source) passed with 128 targets each. Their C++ ordered-count vectors and complete Rust collector records are byte-identical. Each Python replay independently found five useful targets, five verified triples, and rank five. The binaries, commands, receipts, logs, and failed attempts are retained in `graph_execution_20261022/`.

Boundary and next action

The graph-specific runner preserves four phase receipts and requires frozen admission in scientific mode. The fixed controls are outside discovery and confirmation. Bind target roots, check-arm quotas, source hashes, admission and output paths in a runnable Coordinator contract before scientific trials.